

Cyber Forensics - 24CY611

Lab 3 - Data Acquisition and Duplication

Creating dd image file using dd command

sudo dd if=/dev/sda of=image.dd bs=4M status=progress

```
(kali㉿kali)-[/]
$ sudo dd if=/dev/sda of=image.dd bs=4M status=progress
1006632960 bytes (1.0 GB, 960 MiB) copied, 4 s, 252 MB/s^C
241+0 records in
241+0 records out
1010827264 bytes (1.0 GB, 964 MiB) copied, 5.06211 s, 200 MB/s

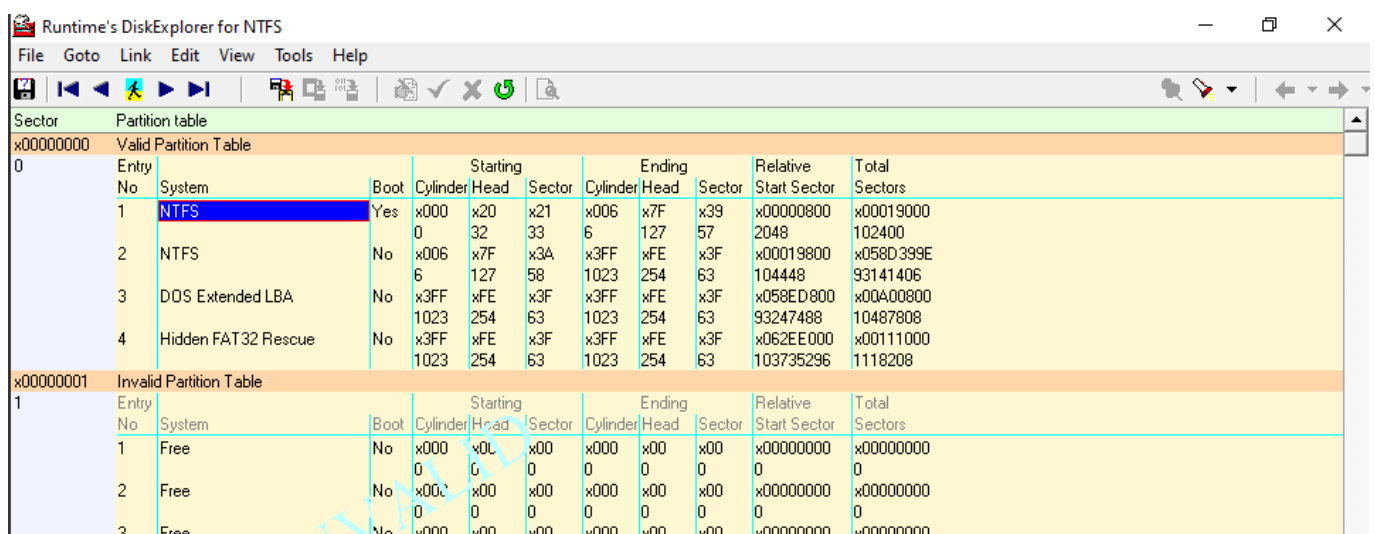
(kali㉿kali)-[/]
$ ls
bin boot dev etc home image.dd initrd.img initrd.img.old lib

(kali㉿kali)-[/]
$
```

1. Investigating NTFS Drive Using DiskExplorer for NTFS

Runtime's DiskExplorer for NTFS is a low-level disk editor that allows users to inspect and manually edit NTFS file system structures. It helps in recovering lost data by accessing raw disk sectors and system metadata directly.

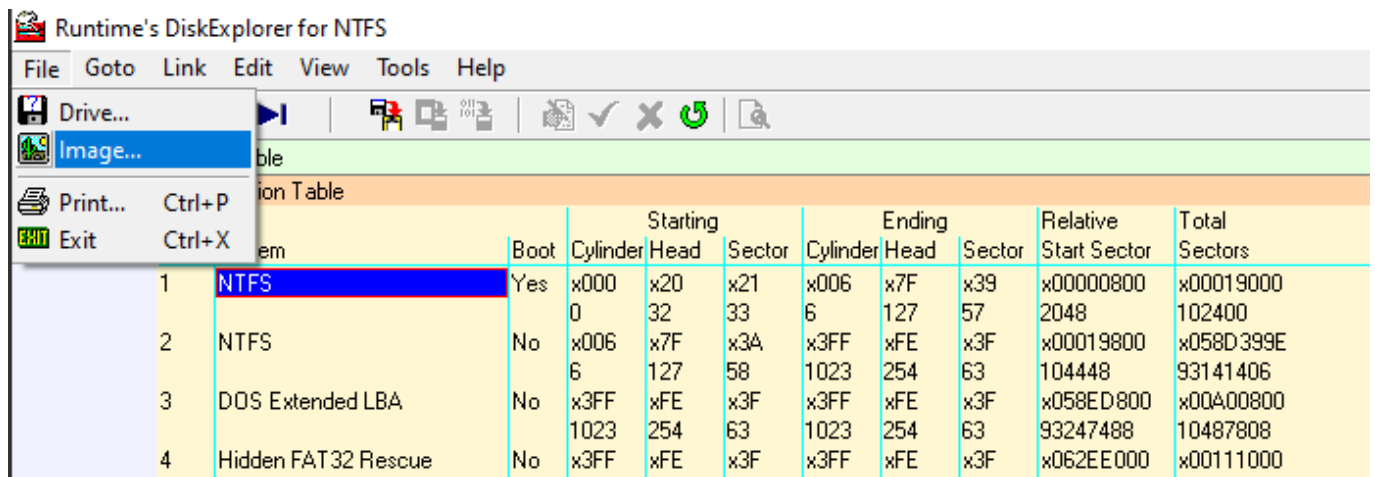
1.Open DiskExplorer for NTFS.



The screenshot shows the 'Runtime's DiskExplorer for NTFS' application window. The 'Partition table' tab is selected, displaying a table of disk partitions. The table has columns for Entry No, System, Boot, Starting (Cylinder, Head, Sector), Ending (Cylinder, Head, Sector), Relative Start Sector, and Total Sectors. Entry 0 is a 'Valid Partition Table' containing four entries: 1 (NTFS, bootable), 2 (NTFS), 3 (DOS Extended LBA), and 4 (Hidden FAT32 Rescue). Entry 1 is an 'Invalid Partition Table' containing three 'Free' entries.

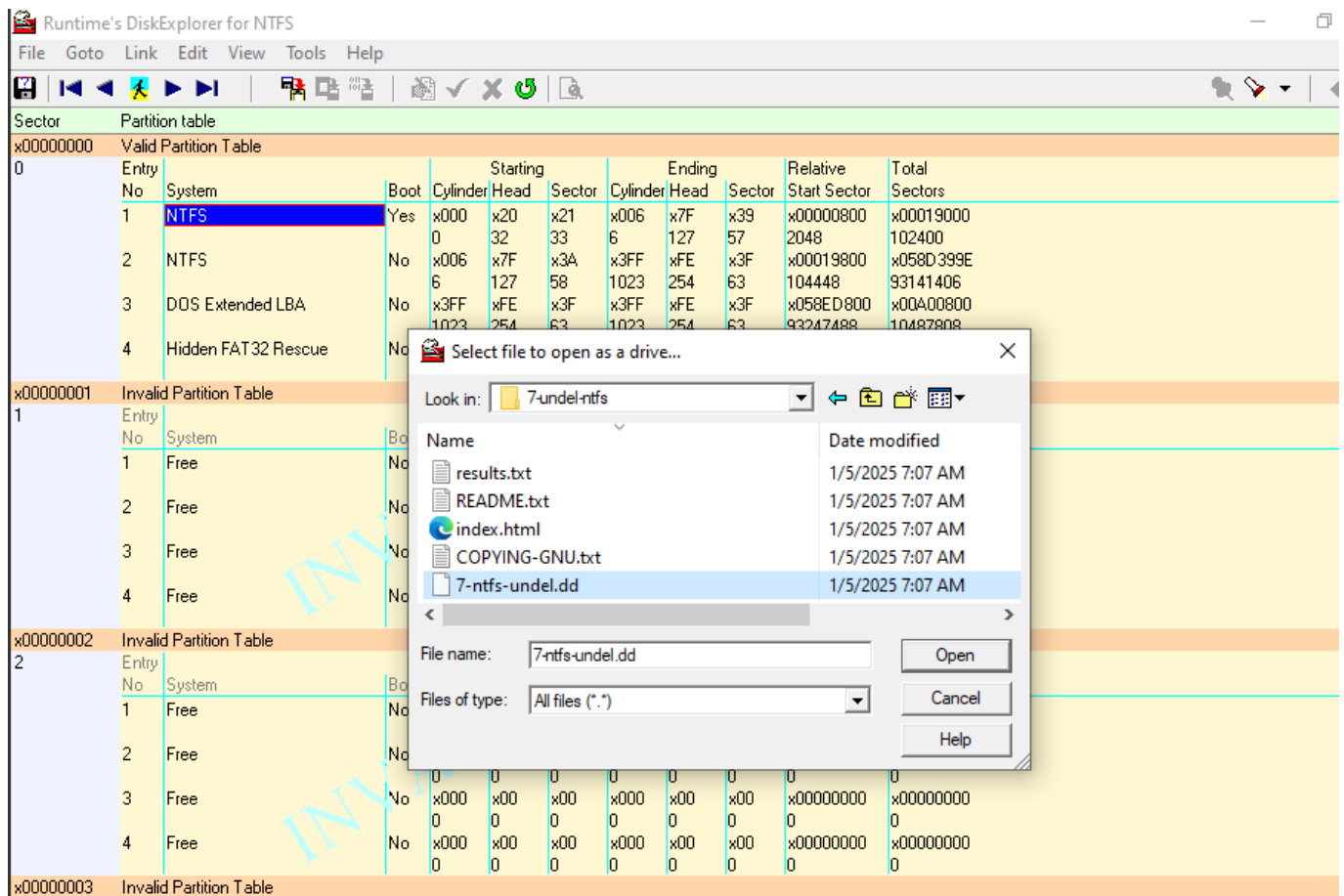
Entry No	System	Boot	Starting Cylinder	Starting Head	Starting Sector	Ending Cylinder	Ending Head	Ending Sector	Relative Start Sector	Total Sectors
Valid Partition Table										
1	NTFS	Yes	x000	x20	x21	x006	x7F	x39	x00000800	x00019000
2	NTFS	No	x006	x7F	x3A	x3FF	xFE	x3F	x00019800	x058D399E
3	DOS Extended LBA	No	x3FF	xFE	x3F	x3FF	xFE	x3F	x058ED800	x00A00800
4	Hidden FAT32 Rescue	No	x3FF	xFE	x3F	x3FF	xFE	x3F	x062EE000	x00111000
Invalid Partition Table										
1	Free	No	x000	x00	x00	x000	x00	x00	x00000000	x00000000
2	Free	No	x000	x00	x00	x000	x00	x00	x00000000	x00000000
3	Free	No	x000	x00	x00	x000	x00	x00	x00000000	x00000000

2. Navigate to files and select images to import the evidence image

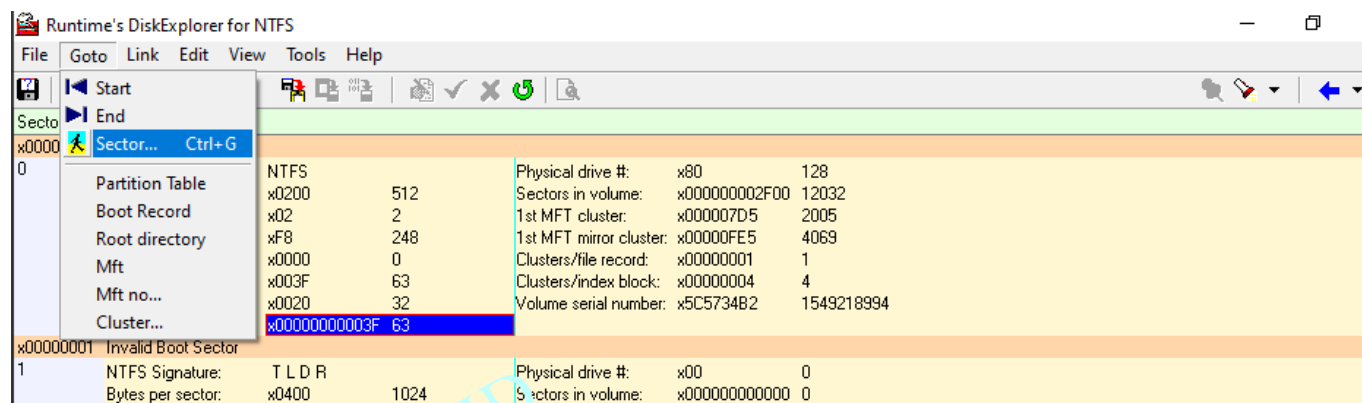


3. Select the evidence image . In the files of type drop down ,select All files and click open.

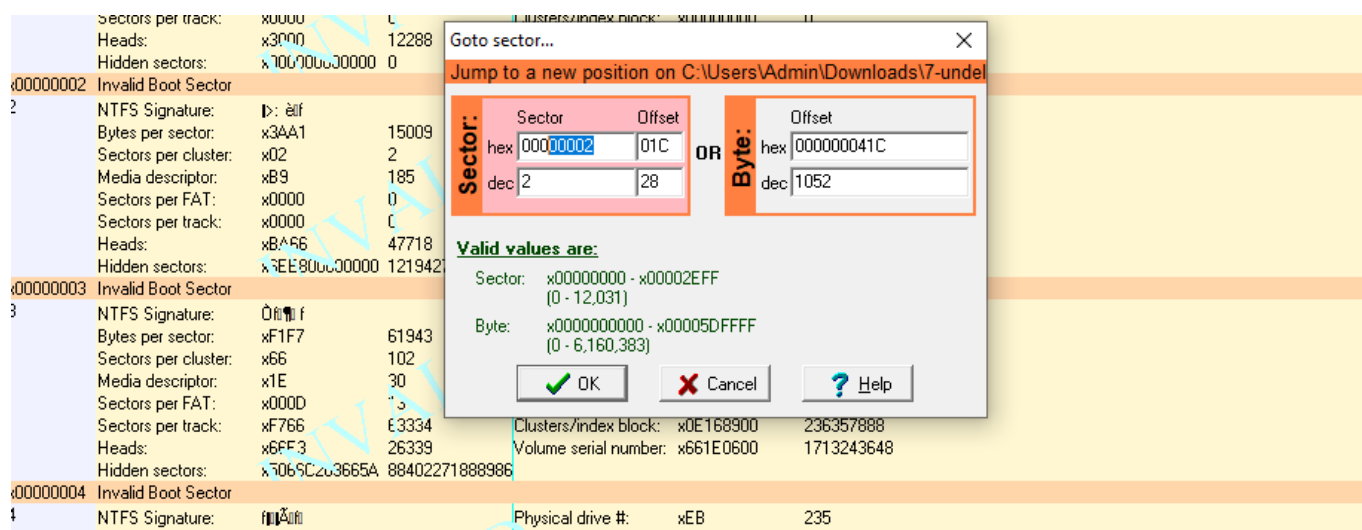
4. It will display the complete details of the image file containing valid and invalid Boot sector.



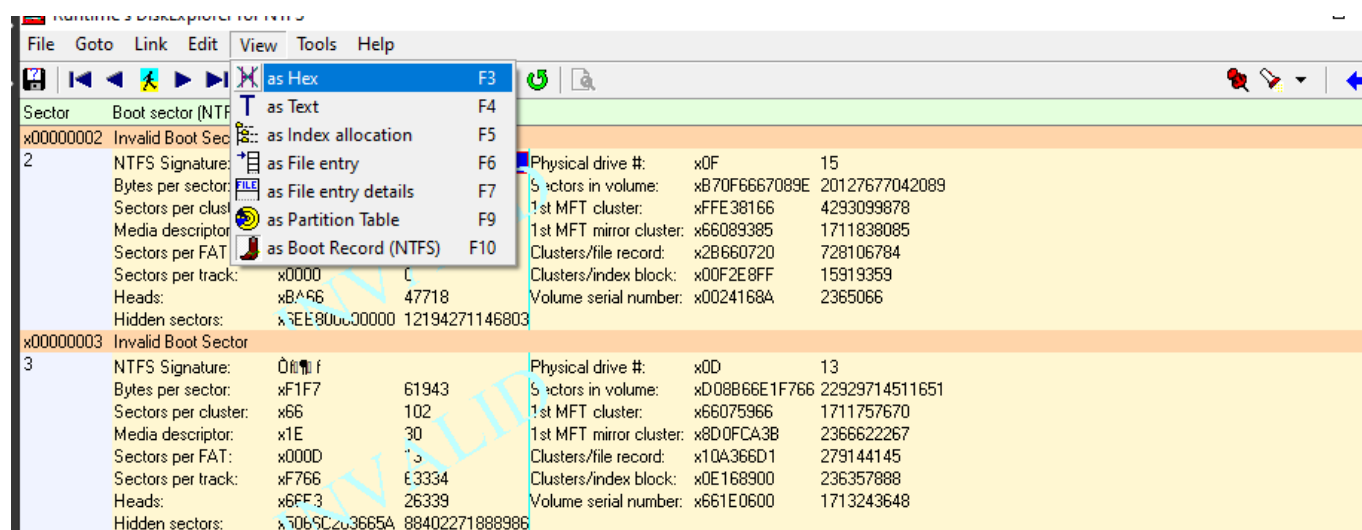
5. Click on Goto and select sector



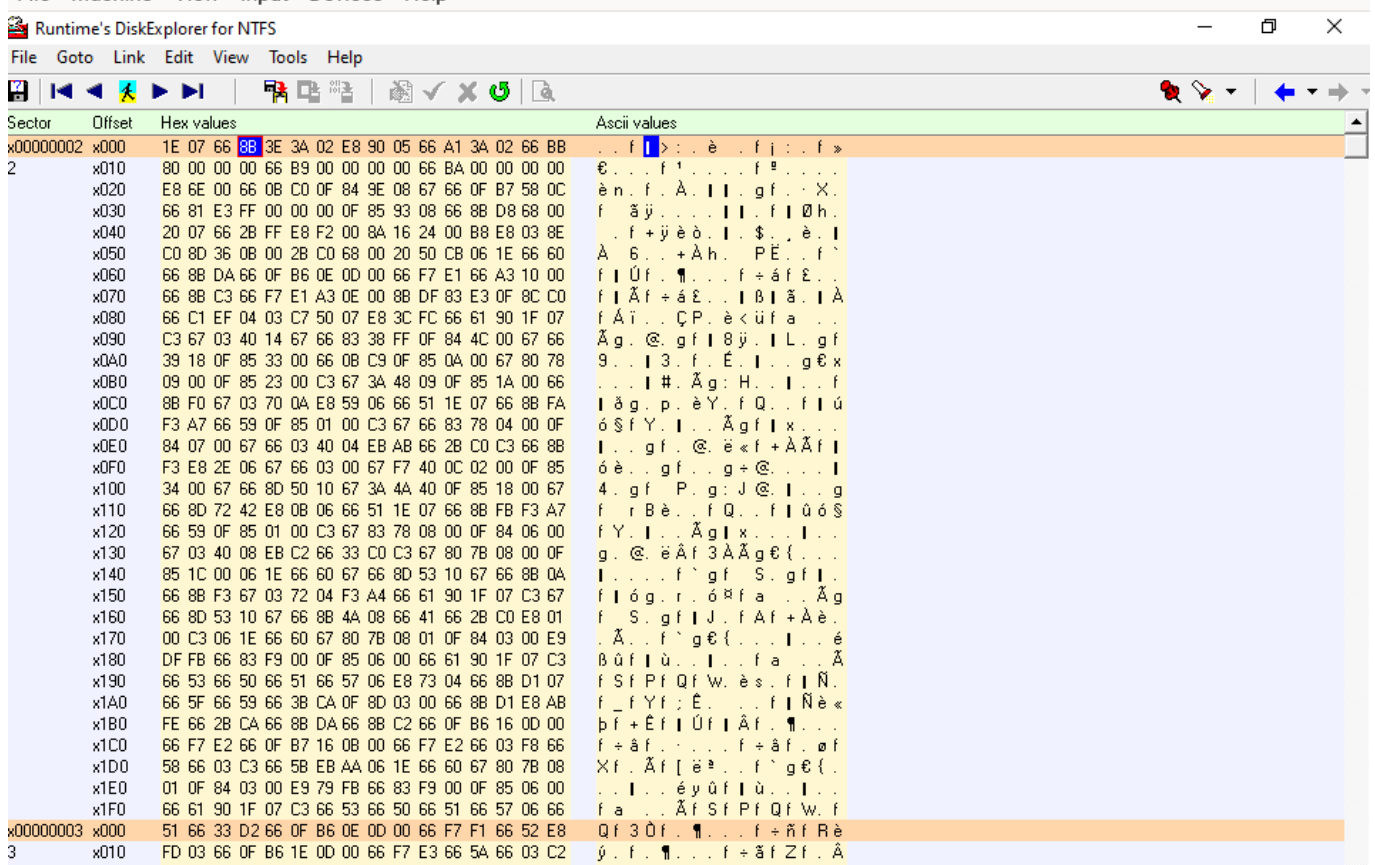
6. Now set the sector to 2 in hex and click ok.



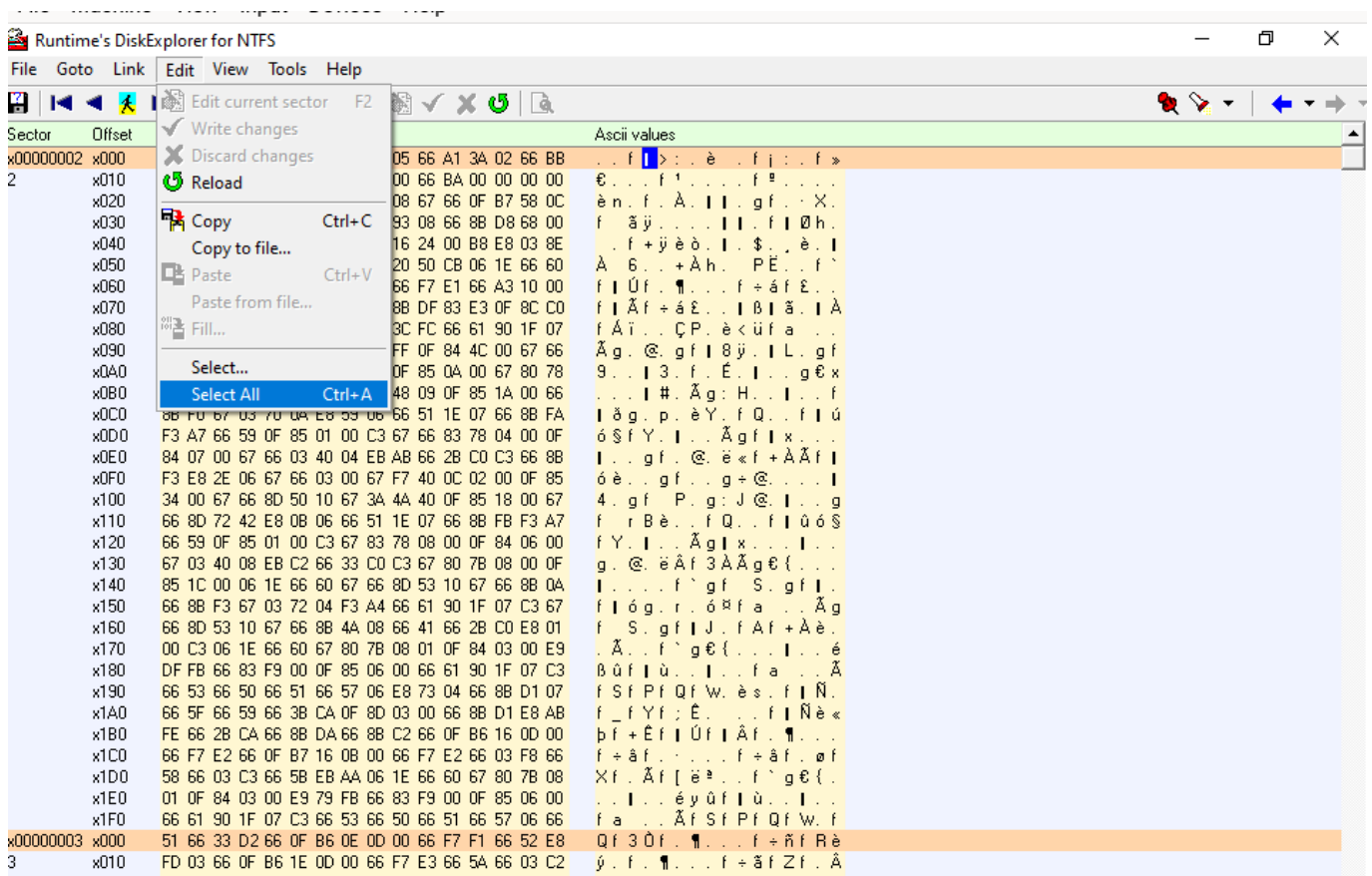
6. In view select as hex.

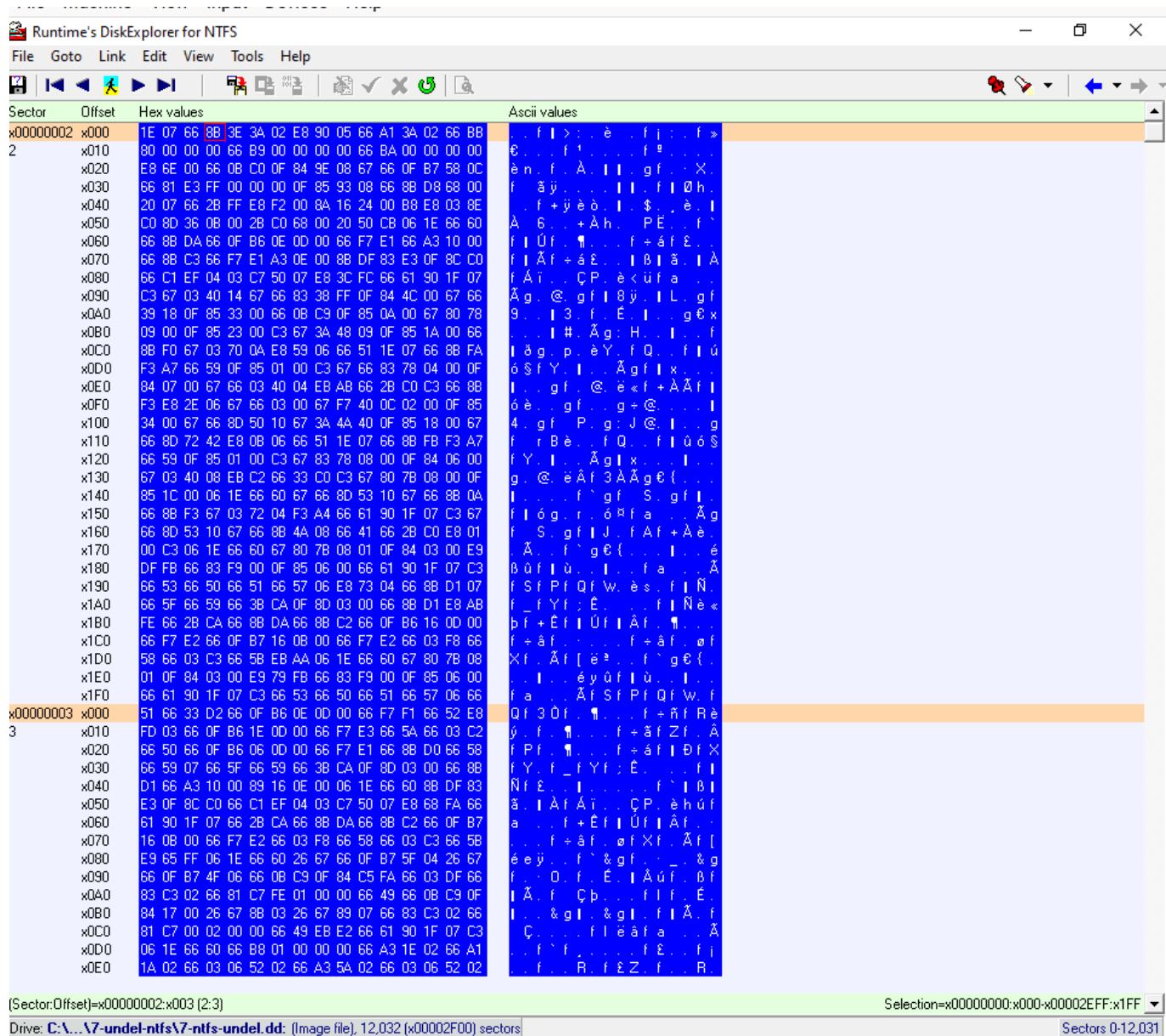


7. It will display the contents as hex values

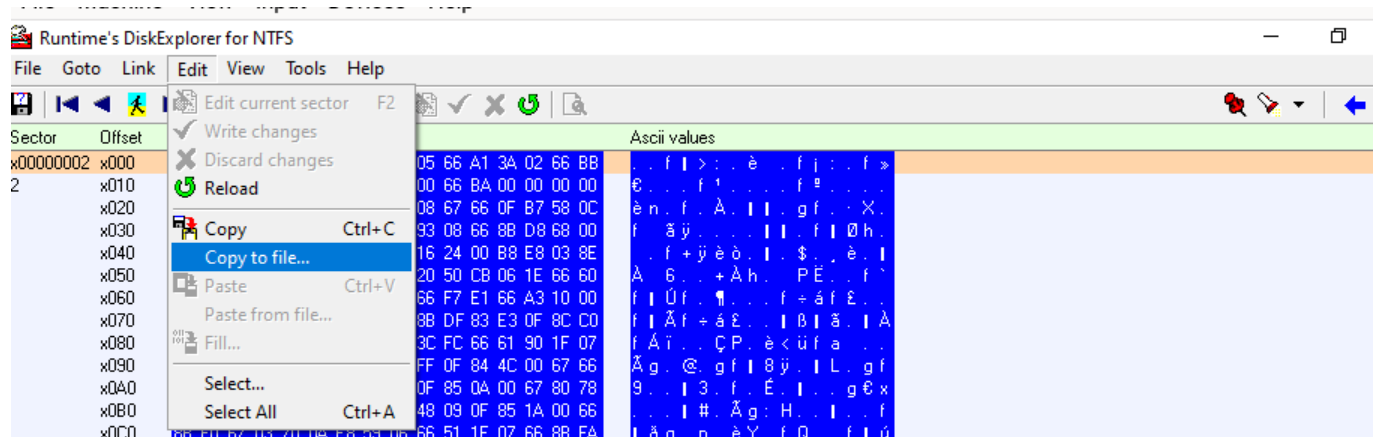


8. Select all from edit

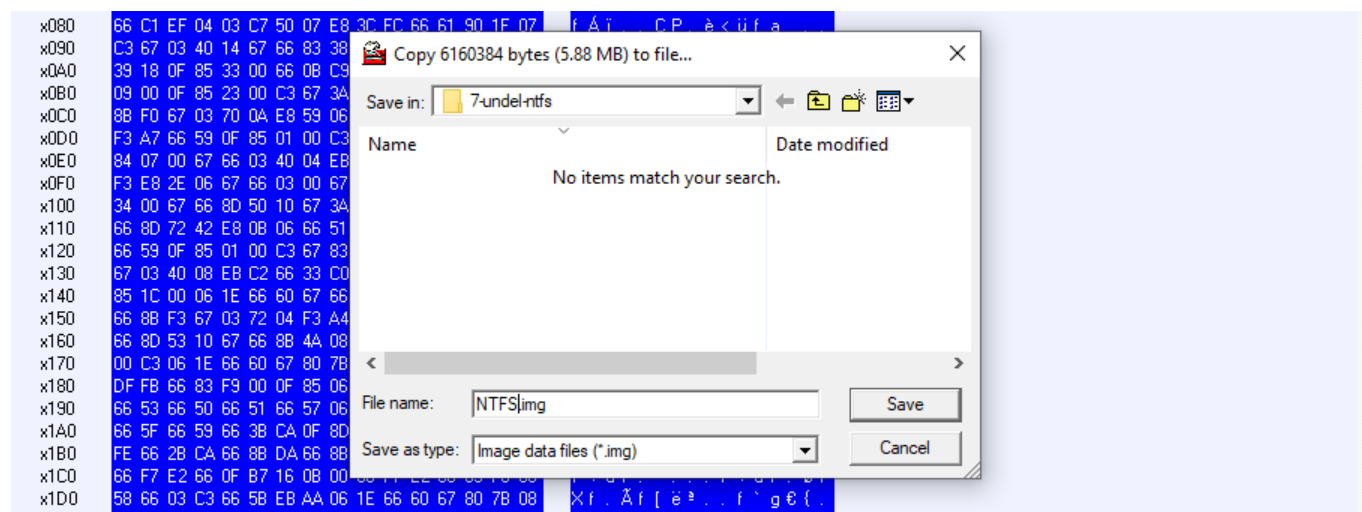




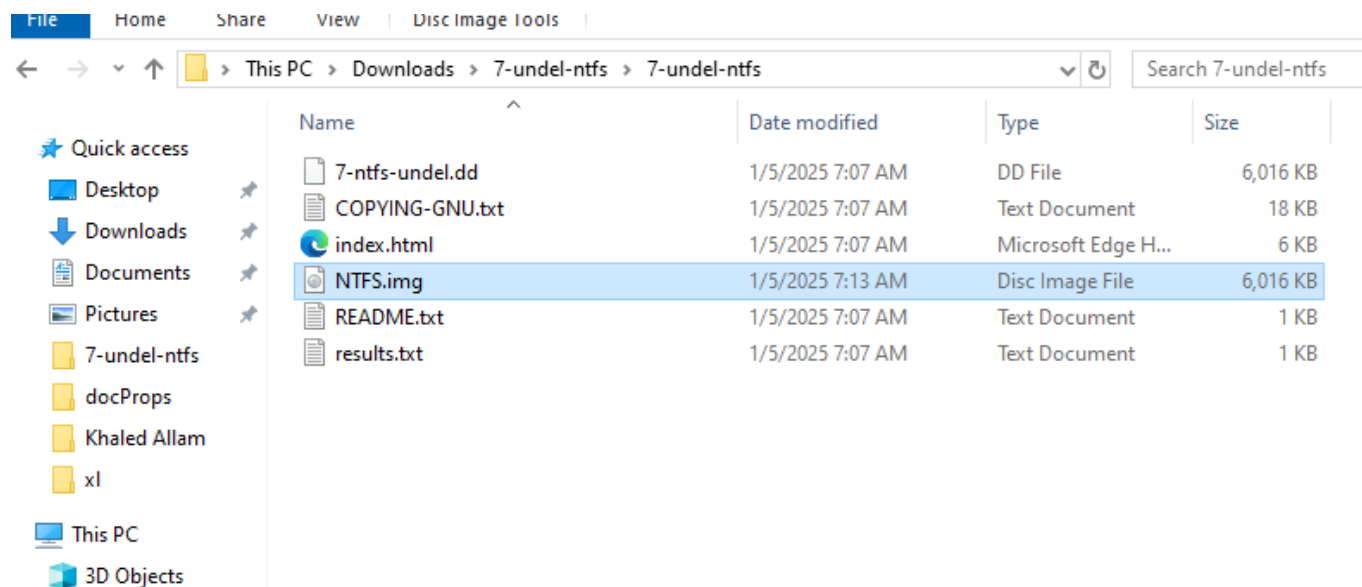
9. Now select copy to files from edit.



10. Then select the folder to save the file



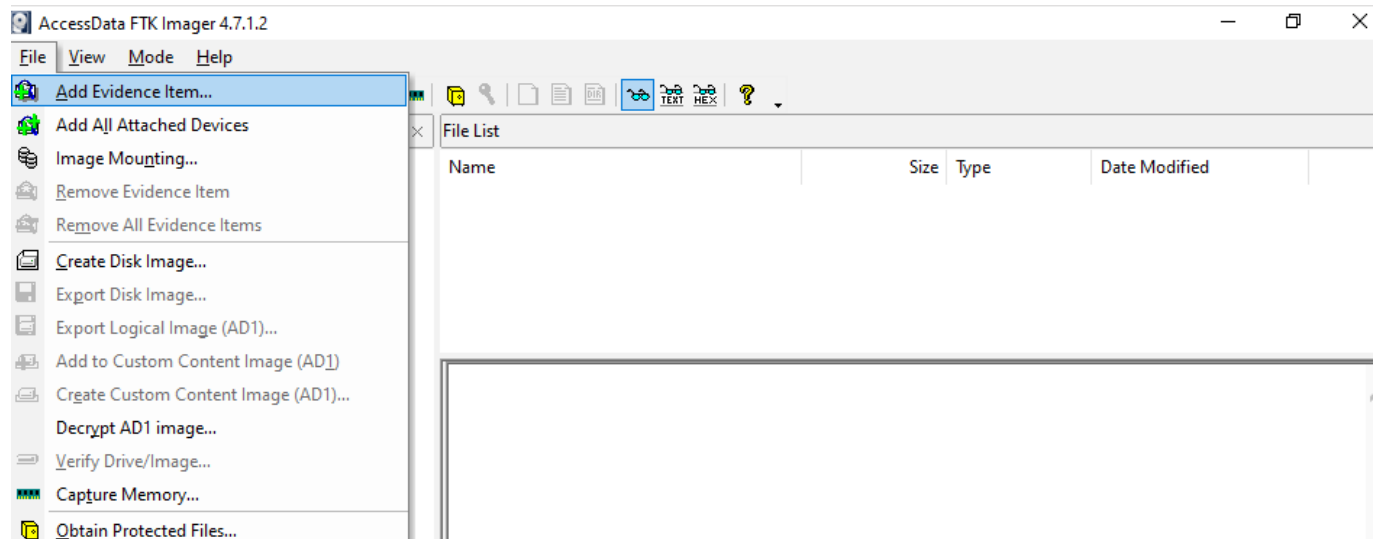
11. The hex values are stored as a img file.



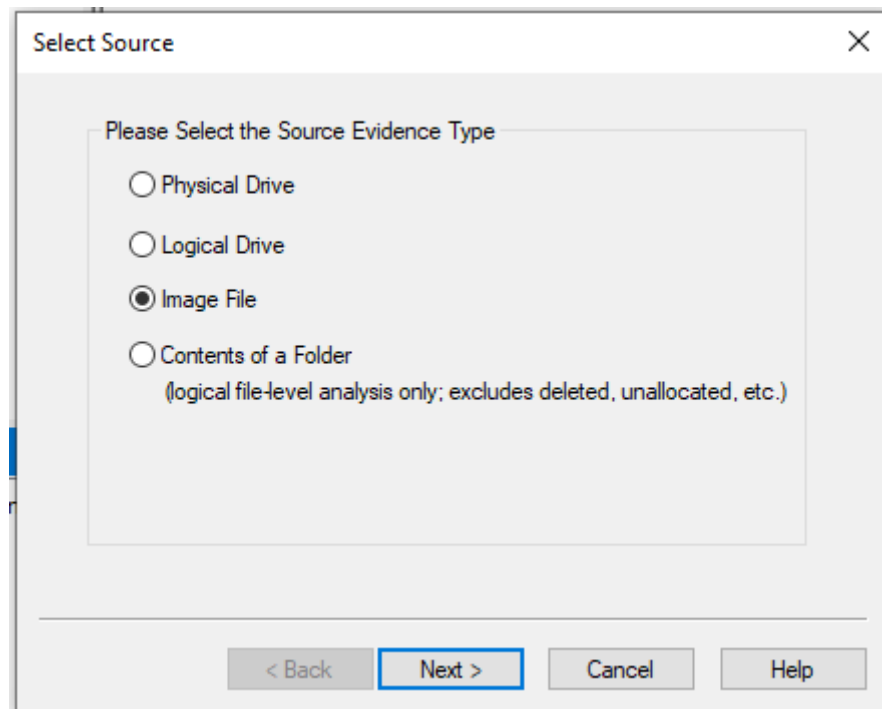
2. Viewing the contents of forensic image using FTK Imager tool

FTK Imager is a forensic tool used to create disk images and access data from drives, partitions, or files. It allows users to preview and recover deleted files, view file system metadata, and export evidence without altering the original data.

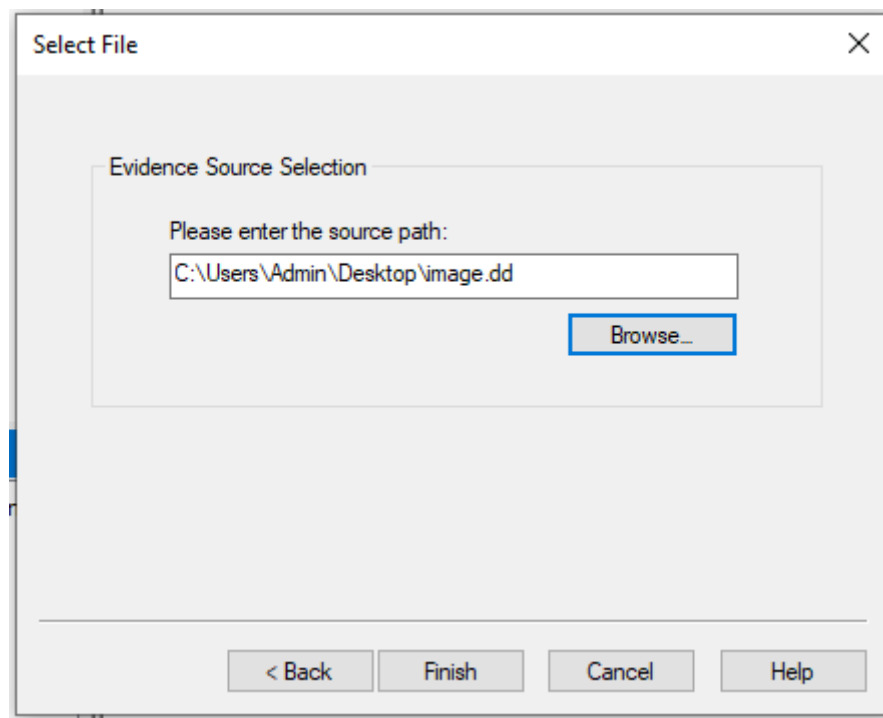
1. Open FTK imager
2. Select Add Evidence Item from the File menu



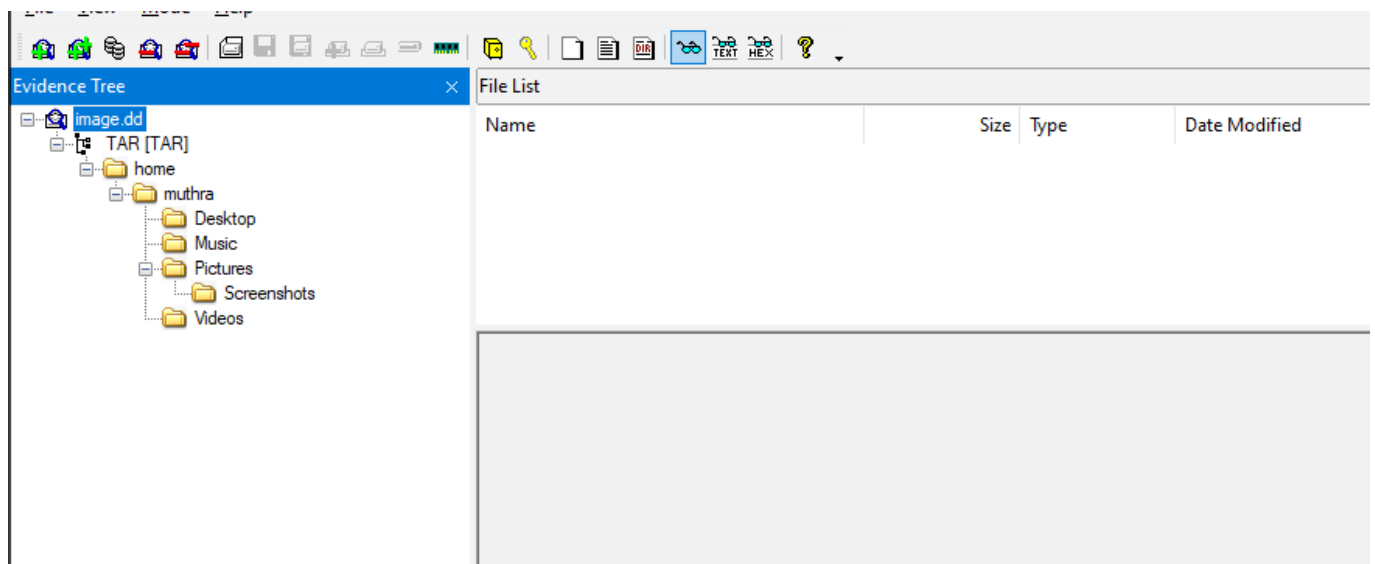
3. Choose Image file from the select source and click next.



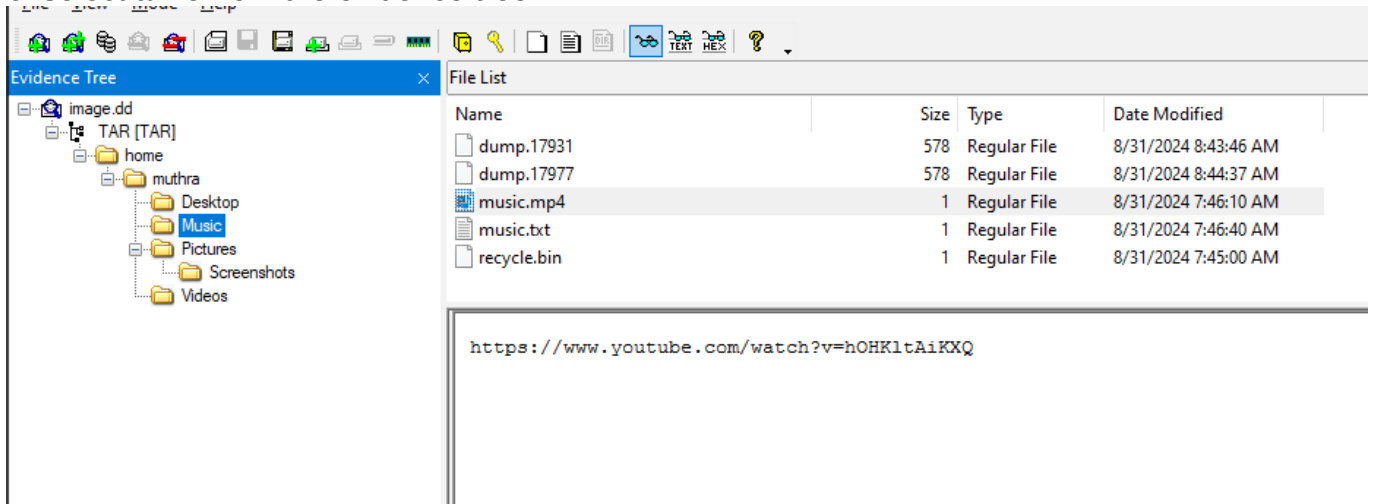
4. Choose the image file using the browse option and click finish



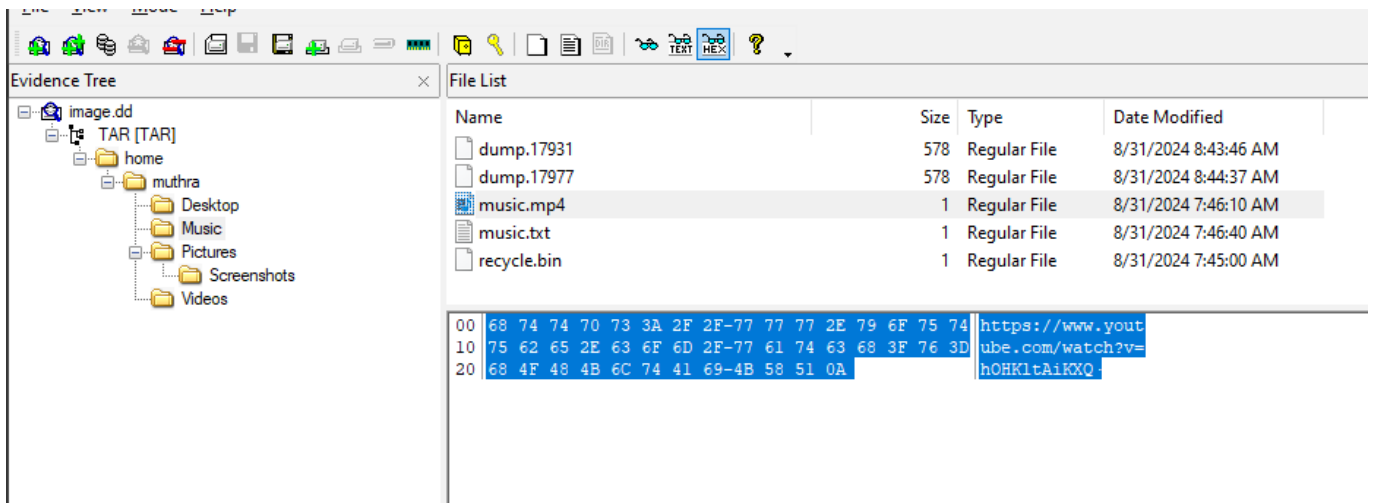
5. Evidence tree will show the contents of the image file



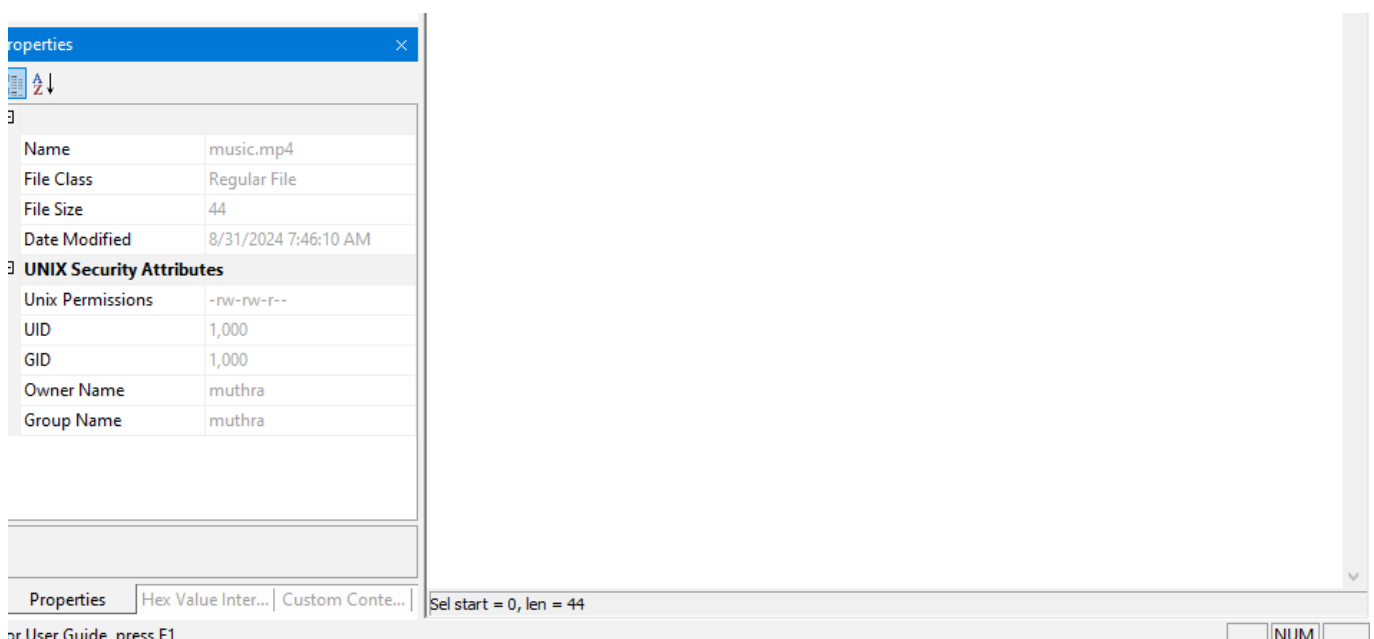
6. select a file from the evidence tree



7. To view the hex value of a particular file , select the file from the File List and click the HEX icon on the toolbar.



8. Now select the properties in the left lower panel to view the properties such as uid ,gid, date, group , etc,...



9. Now select the Hex value interpreter in the left lower panel to view the properties such as signed integer , DOS , time, unsigned integer etc,.

